

Unitree Week 9

Presentation by Mario Ortiz

This week

- Now that I am able to use an environment on Isaac Lab + Isaac Sim that allows me to train the G1 robot, I will try to make my own G1 code.
- Specific goals are:
 - Training the Unitree G1 to walk up stairs
 - Training the Unitree G1 to walk while holding items / training unitree to pick up items.

Training G1 to walk stairs

After further research into the IsaacLab directory, I found this file: “C:\Users\Mario\IsaacLab\source\isaacsim\isaacsim\terrains\trimesh\mesh_terrains.py”, containing various functions regarding the terrain. In this code, there is a pyramid terrain.

```
& "C:\Users\Mario\isaac-sim\kit\python\python.exe"  
source/standalone/workflows/rsl_rl/train.py --task  
Isaac-Velocity-Stairs-G1-v0 --num_envs 1024 --headless
```

Training G1 to walk stairs

I will now edit the `rough_env_cfg.py` code so that the terrain is a pyramid terrain instead of a rough terrain (will force the g1 to learn to walk up stairs).

stair_env_cfg.py

```
C:\Users\Mario> IsaacLab > source > isaacsim > isaacsim_tasks > manager_based > locomotion > velocity > config > g1 > stair_env_cfg.py > G1StairRewards

1 # Copyright (c) 2022-2025, The Isaac Lab Project Developers.
2 # SPDX-License-Identifier: BSD-3-Clause
3
4 from isaacsim.managers import RewardTermCfg as RewardTerm
5 from isaacsim.managers import SceneEntityCfg
6 from isaacsim.utils import configclass
7
8 from isaacsim.terrains import MeshPlaneTerrainCfg, MeshPyramidStairsTerrainCfg
9 import isaacsim_tasks.manager_based.locomotion.velocity.mdp as mdp
10 from isaacsim_tasks.manager_based.locomotion.velocity.velocity_env_cfg import LocomotionVelocityRoughEnvCfg, RewardsCfg
11
12 ##
13 # Pre-defined configs
14 ##
15 from isaacsim_assets import G1_MINIMAL_CFG
16
17 @configclass
18 class G1StairRewards(RewardsCfg):
19     """Reward terms mirrored from the working G1 Rough config."""
20
21     termination_penalty = RewardTerm(func=mdp.is_terminated, weight=-200.0)
22     track_lin_vel_xy_exp = RewardTerm(
23         func=mdp.track_lin_vel_xy_world_exp,
24         weight=1.0,
25         params={"command_name": "base_velocity", "std": 0.5},
26     )
27     track_ang_vel_z_exp = RewardTerm(
28         func=mdp.track_ang_vel_z_world_exp,
29         weight=2.0,
30         params={"command_name": "base_velocity", "std": 0.5},
31     )
32     feet_air_time = RewardTerm(
33         func=mdp.feet_air_time_positive_biped,
34         weight=0.25,
35         params={
36             "command_name": "base_velocity",
37             "sensor_cfg": SceneEntityCfg("contact_forces", body_names=".*ankle_roll_link"),
38             "threshold": 0.4,
39         },
40     )
41     feet_slide = RewardTerm(
42         func=mdp.feet_slide,
43         weight=-0.1,
44         params={
45             "sensor_cfg": SceneEntityCfg("contact_forces", body_names=".*ankle_roll_link"),
46             "asset_cfg": SceneEntityCfg("robot", body_names=".*ankle_roll_link"),
47         },
48     )
49     # Penalties
50     dof_pos_limits = RewardTerm(
51         func=mdp.joint_pos_limits,
52         weight=-1.0,
53         params={"asset_cfg": SceneEntityCfg("robot", joint_names=[".*ankle_pitch_joint", ".*ankle_roll_joint"])},
54     )
55     joint_deviation Arms = RewardTerm(
```

- Added library imports related to Pyramid terrain
- Simply renamed class to stair environment
- Reduced the reward terms to focus on training the robot to walk up/down steps.
- Implemented a structured pyramid terrain where 80% of the map contains pyramid terrain and 20% of the map contains flat terrain.
- Added more constraints to the robot's movements to ensure it is rewarded for moving directly forward (pose_range, velocity_range)

Stair_env_cfg.py (continued)

```
55 joint_deviation_arms = NewTerm(
56     funt=mdp.joint_deviation_l1,
57     weights=0.1,
58     params={
59         "asset_cfg": SceneEntityCfg(
60             "robot",
61             joint_names=["*_shoulder_*", "*_elbow_*"],
62         ),
63     },
64 )
65
66 @configclass
67 class G1StairEnvCfg(LocomotionVelocityRoughEnvCfg):
68     rewards = G1StairRewards()
69
70     def __post_init__(self):
71         super().__post_init__()
72
73         # Scene and Asset setup
74         self.scene.robot = G1_MINIMAL_CFG.replace(prim_path="{ENV_REGEX_NS}/Robot")
75         self.scene.height_scanner.prim_path = "{ENV_REGEX_NS}/Robot/torso_link"
76
77         # Randomization and events
78         self.events.push_robot = None
79         self.events.add_base_mass = None
80         self.events.base_com = None
81         self.events.reset_robot_joints.params["position_range"] = (1.0, 1.0)
82         self.events.base_external_force_torque.params["asset_cfg"].body_names = ["torso_link"]
83
84         # Reset pose
85         self.events.reset_base.params = {
86             "pose_range": {"x": (-0.2, 0.2), "y": (-0.2, 0.2), "yaw": (-3.14, 3.14)},
87             "velocity_range": {k: (0.0, 0.0) for k in ["x", "y", "z", "roll", "pitch", "yaw"]},
88         }
89
90         # Rewards
91         self.rewards.lin_vel_z_l2.weight = 0.0
92         self.rewards.undesired_contacts = None
93         self.rewards.flat_orientation_l2.weight = -1.0
94         self.rewards.action_rate_l2.weight = +0.005
95         self.rewards.dof_acc_l2.weight = -1.25e-7
96         self.rewards.dof_torques_l2.weight = -1.5e-7
97
98         # Terminations
99         self.terminations.base_contact.params["sensor_cfg"].body_names = "torso_link"
100
101         # Movement commands -> forward moving
102         self.commands.base_velocity.ranges.lin_vel_x = (0.0, 1.0)
103         self.commands.base_velocity.ranges.lin_vel_y = (0.0, 0.0)
104         self.commands.base_velocity.ranges.ang_vel_z = (-0.5, 0.5)
105
106         # Adding the pyramid terrain
107         if self.scene.terrain.terrain_generator is not None:
108             stairs_cfg = MeshPyramidStairsTerrainCfg()
109             flat_cfg = MeshPlaneTerrainCfg()
```

```
110         stairs_cfg = MeshPyramidStairsTerrainCfg()
111         stairs_cfg.proportion = 0.8
112         stairs_cfg.step_width = 0.3
113         stairs_cfg.step_height_range = (0.05, 0.1)
114         stairs_cfg.platform_width = 1.0
115         stairs_cfg.border_width = 0.0
116         stairs_cfg.holes = False
117
118         flat_cfg.proportion = 0.2
119
120         self.scene.terrain.terrain_generator.sub_terrains = {
121             "pyramid_stairs": stairs_cfg,
122             "flat": flat_cfg
123         }
124
125 @configclass
126 class G1StairEnvCfg_PLAY(G1StairEnvCfg):
127     def __post_init__(self):
128         super().__post_init__()
129         self.scene.num_envs = 32
130         self.commands.base_velocity.ranges.lin_vel_x = (0.5, 0.5)
131         self.commands.base_velocity.ranges.lin_vel_y = (0.0, 0.0)
132         self.commands.base_velocity.ranges.ang_vel_z = (0.0, 0.0)
```

First Attempt Logs

```
#####
Learning iteration 62/6000

Computation: 13948 steps/s (collection: 3.334s, learning 0.190s)
  Mean action noise std: 1.01
  Mean value_function loss: 0.0130
  Mean surrogate loss: -0.0156
  Mean entropy loss: 52.3281
  Mean reward: -4.89
  Mean episode length: 48.50
Episode_Reward/track_lin_vel_xy_exp: 0.0141
Episode_Reward/track_ang_vel_z_exp: 0.0065
  Episode_Reward/lin_vel_z_l2: 0.0000
  Episode_Reward/ang_vel_xy_l2: -0.0201
  Episode_Reward/dof_torques_l2: -0.0002
  Episode_Reward/dof_acc_l2: -0.0114
  Episode_Reward/action_rate_l2: -0.0183
  Episode_Reward/feet_air_time: 0.0001
Episode_Reward/flat_orientation_l2: -0.0042
  Episode_Reward/dof_pos_limits: -0.0005
Episode_Reward/termination_penalty: -0.2000
  Episode_Reward/feet_slide: -0.0041
Episode_Reward/joint_deviation_arms: -0.0040
  Curriculum/terrain_levels: 0.0000
Metrics/base_velocity/error_vel_xy: 0.0003
Metrics/base_velocity/error_vel_yaw: 0.4059
  Episode_Termination/time_out: 0.0000
  Episode_Termination/base_contact: 1.0000
-----
Total timesteps: 3096576
Iteration time: 3.52s
Time elapsed: 00:04:05
ETA: 06:26:00
```

First Attempt Logs

```
#####
Learning iteration 317/6000

      Computation: 11358 steps/s (collection: 4.071s, learning 0.257s)
      Mean action noise std: 1.15
      Mean value function loss: 0.0206
      Mean surrogate loss: -0.0124
      Mean entropy loss: 53.7381
      Mean reward: -8.90
      Mean episode length: 768.78
Episode_Reward/track_lin_vel_xy_exp: 0.3843
Episode_Reward/track_ang_vel_z_exp: 0.1842
      Episode_Reward/lin_vel_z_l2: 0.0000
      Episode_Reward/ang_vel_xy_l2: -0.1292
      Episode_Reward/dof_torques_l2: -0.0023
      Episode_Reward/dof_acc_l2: -0.1799
      Episode_Reward/action_rate_l2: -0.4436
      Episode_Reward/feet_air_time: 0.0048
Episode_Reward/flat_orientation_l2: -0.0371
      Episode_Reward/dof_pos_limits: -0.0164
Episode_Reward/termination_penalty: -0.0580
      Episode_Reward/feet_slide: -0.0697
Episode_Reward/joint_deviation_arms: -0.1060
      Curriculum/terrain_levels: 4.1235
Metrics/base_velocity/error_vel_xy: 0.8602
Metrics/base_velocity/error_vel_yaw: 4.5335
      Episode_Termination/time_out: 0.5913
      Episode_Termination/base_contact: 0.4087
-----
      Total timesteps: 15630336
      Iteration time: 4.33s
      Time elapsed: 00:20:26
      ETA: 06:05:15
```

First Attempt Logs

```
#####
Learning iteration 424/6000

      Computation: 13119 steps/s (collection: 3.539s, learning 0.207s)
      Mean action noise std: 1.17
      Mean value_function loss: 0.0152
      Mean surrogate loss: -0.0119
      Mean entropy loss: 53.3059
      Mean reward: -4.23
      Mean episode length: 909.24
Episode_Reward/track_lin_vel_xy_exp: 0.5473
Episode_Reward/track_ang_vel_z_exp: 0.2574
      Episode_Reward/lin_vel_z_l2: 0.0000
      Episode_Reward/ang_vel_xy_l2: -0.0996
      Episode_Reward/dof_torques_l2: -0.0022
      Episode_Reward/dof_acc_l2: -0.1587
      Episode_Reward/action_rate_l2: -0.5076
      Episode_Reward/feet_air_time: 0.0062
Episode_Reward/flat_orientation_l2: -0.0248
      Episode_Reward/dof_pos_limits: -0.0176
Episode_Reward/termination_penalty: -0.0243
      Episode_Reward/feet_slide: -0.0658
Episode_Reward/joint_deviation_arms: -0.1179
      Curriculum/terrain_levels: 5.6663
Metrics/base_velocity/error_vel_xy: 0.7198
Metrics/base_velocity/error_vel_yaw: 3.9430
      Episode_Termination/time_out: 0.8060
      Episode_Termination/base_contact: 0.1940
-----
      Total timesteps: 20889600
      Iteration time: 3.75s
      Time elapsed: 00:27:47
      ETA: 06:04:40
```

First Attempt Logs

```
#####
Learning iteration 625/6000

      Computation: 12890 steps/s (collection: 3.605s, learning 0.208s)
      Mean action noise std: 1.19
      Mean value_function loss: 0.0103
      Mean surrogate loss: -0.0098
      Mean entropy loss: 51.3874
      Mean reward: 2.69
      Mean episode length: 981.33
Episode_Reward/track_lin_vel_xy_exp: 0.6983
Episode_Reward/track_ang_vel_z_exp: 0.4132
      Episode_Reward/lin_vel_z_l2: 0.0000
      Episode_Reward/ang_vel_xy_l2: -0.0661
      Episode_Reward/dof_torques_l2: -0.0018
      Episode_Reward/dof_acc_l2: -0.1268
      Episode_Reward/action_rate_l2: -0.5442
      Episode_Reward/feet_air_time: 0.0061
Episode_Reward/flat_orientation_l2: -0.0201
      Episode_Reward/dof_pos_limits: -0.0158
Episode_Reward/termination_penalty: -0.0028
      Episode_Reward/feet_slide: -0.0557
Episode_Reward/joint_deviation_arms: -0.1278
      Curriculum/terrain_levels: 5.9219
Metrics/base_velocity/error_vel_xy: 0.5842
Metrics/base_velocity/error_vel_yaw: 2.7130
      Episode_Termination/time_out: 0.9071
      Episode_Termination/base_contact: 0.0929
-----
      Total timesteps: 30769152
      Iteration time: 3.81s
      Time elapsed: 00:39:58
      ETA: 05:43:16
```

First Attempt Logs

```
#####
Learning iteration 2811/6000

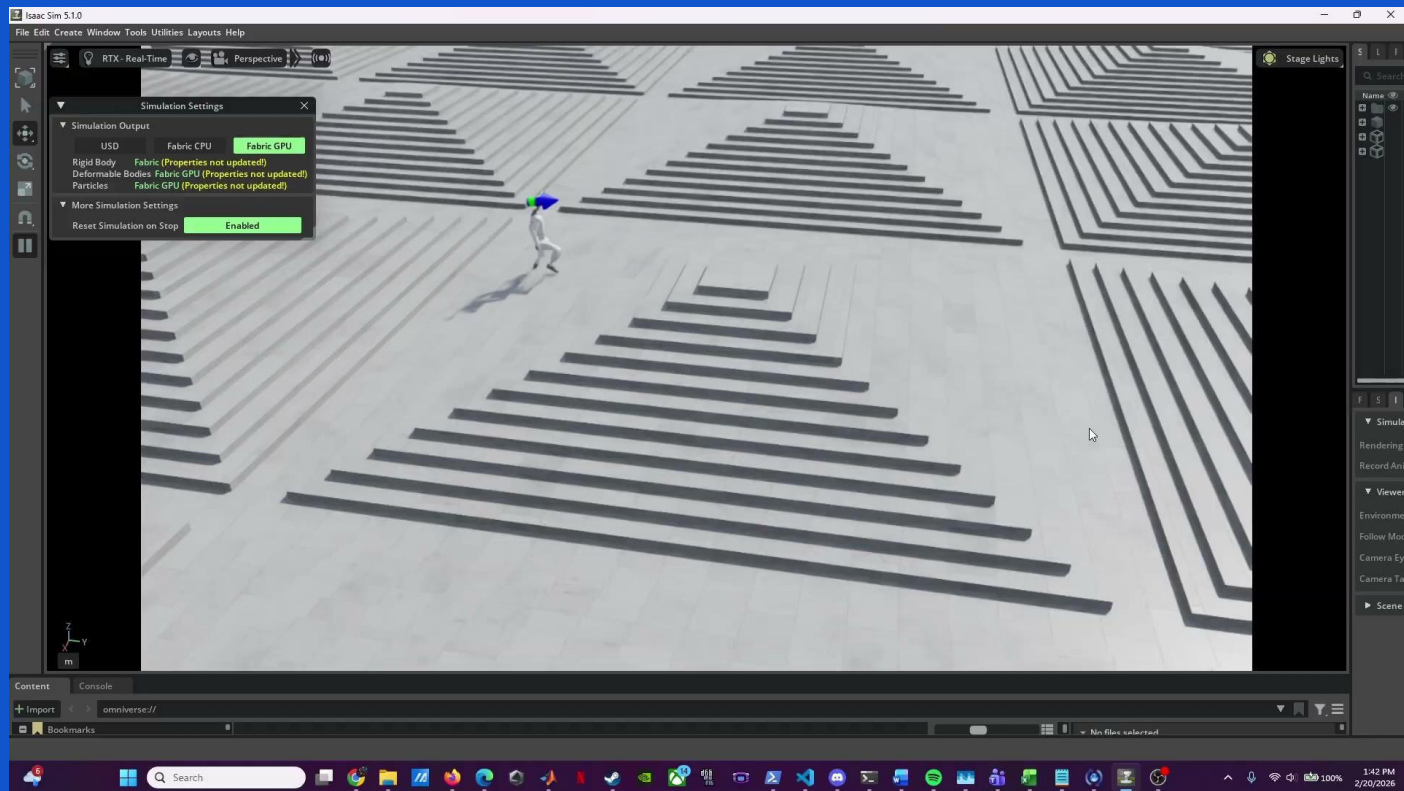
      Computation: 13523 steps/s (collection: 3.447s, learning 0.188s)
      Mean action noise std: 1.29
      Mean value_function loss: 0.0114
      Mean surrogate loss: -0.0067
      Mean entropy loss: 49.2000
      Mean reward: 16.94
      Mean episode length: 941.53
Episode_Reward/track_lin_vel_xy_exp: 0.7377
Episode_Reward/track_ang_vel_z_exp: 1.0495
      Episode_Reward/lin_vel_z_l2: 0.0000
      Episode_Reward/ang_vel_xy_l2: -0.0379
      Episode_Reward/dof_torques_l2: -0.0027
      Episode_Reward/dof_acc_l2: -0.0889
      Episode_Reward/action_rate_l2: -0.5734
      Episode_Reward/feet_air_time: 0.0057
Episode_Reward/flat_orientation_l2: -0.0437
      Episode_Reward/dof_pos_limits: -0.0259
Episode_Reward/termination_penalty: -0.0328
      Episode_Reward/feet_slide: -0.0456
Episode_Reward/joint_deviation_arms: -0.1282
      Curriculum/terrain_levels: 5.8271
Metrics/base_velocity/error_vel_xy: 0.4103
Metrics/base_velocity/error_vel_yaw: 0.7610
      Episode_Termination/time_out: 0.9118
      Episode_Termination/base_contact: 0.0882
-----
      Total timesteps: 138215424
      Iteration time: 3.63s
      Time elapsed: 03:01:57
      ETA: 03:26:20
```

First Attempt Logs

```
#####
Learning iteration 5999/6000

Computation: 13048 steps/s (collection: 3.564s, learning 0.203s)
  Mean action noise std: 1.29
  Mean value_function loss: 0.0121
  Mean surrogate loss: -0.0062
  Mean entropy loss: 47.2697
  Mean reward: 24.58
  Mean episode length: 981.14
Episode_Reward/track_lin_vel_xy_exp: 0.8347
Episode_Reward/track_ang_vel_z_exp: 1.3239
  Episode_Reward/lin_vel_z_l2: 0.0000
  Episode_Reward/ang_vel_xy_l2: -0.0346
  Episode_Reward/dof_torques_l2: -0.0067
  Episode_Reward/dof_acc_l2: -0.0638
  Episode_Reward/action_rate_l2: -0.5954
  Episode_Reward/feet_air_time: 0.0056
Episode_Reward/flat_orientation_l2: -0.0163
  Episode_Reward/dof_pos_limits: -0.0341
Episode_Reward/termination_penalty: -0.0097
  Episode_Reward/feet_slide: -0.0471
Episode_Reward/joint_deviation_arms: -0.1359
  Curriculum/terrain_levels: 5.8658
Metrics/base_velocity/error_vel_xy: 0.3691
Metrics/base_velocity/error_vel_yaw: 0.6185
  Episode_Termination/time_out: 0.9621
  Episode_Termination/base_contact: 0.0379
-----
Total timesteps: 294912000
Iteration time: 3.77s
Time elapsed: 06:22:44
ETA: 00:00:03
```

```
.\isaac.bat -p scripts/reinforcement_learning/rsl_rl/play.py --task  
Isaac-Velocity-Stairs-G1-Play-v0 --checkpoint  
"C:\Users\Mario\IsaacLab\logs\rsl_rl\g1_stairs\2026-02-17_13-20-13\model_5999.pt"
```



Analysis

- The g1 robot has been successfully trained to walk up and down pyramid stairs. However, some issues remain:
 - The technique the robot learned to walk up the stairs: they spread their legs too wide. The robot probably learned that the most effective way to walk up/down the stairs was to lower its center of gravity and widen its stance to ensure stability, reducing the risk of falling.
- To improve this, the RL policy needs to be adjusted; The reward should penalize overly wide stances and encourage stair locomotion patterns that are closer to human walking.
 - The robot can be rewarded for maintaining appropriate leg spacing and stable joint trajectories during each step.

Analysis (continued)

- Another issue stems from the g1 robot not walking straight up the stairs all the time.
 - Instead of walking to the top of the stairs, it may climb it sideways. For our scenario of walking up stairs in the Grove School building, the g1 needs to learn to walk up straight stairs with walls/railings in between them, forcing it to walk straight up.
 - Therefore in the future, will need to make a very specific terrain that mimics that of building staircases.

New code based on analysis

```
66 # Penalties for hip roll and yaw deviation
67 joint_deviation_legs = RewTerm(
68     func=mdp.joint_deviation_l1,
69     weight=-0.2,
70     params={
71         "asset_cfg": SceneEntityCfg(
72             "robot",
73             joint_names=["*_hip_roll_joint", "*_hip_yaw_joint"],
74         )
75     },
76 )
77 # Penalties for squatting (forces robot to stand upright)
78 base_height_l2 = RewTerm(
79     func=mdp.base_height_l2,
80     weight=-2.0,
81     params={"target_height": 0.75, "asset_cfg": SceneEntityCfg("robot")},
82 )
```

Adding new penalties for hip roll and yaw deviations and squatting.

Second Attempt Logs

```
#####  
Learning iteration 231/6000
```

```
Computation: 12829 steps/s (collection: 3.615s, learning 0.216s)
```

```
Mean action noise std: 1.02
```

```
Mean value function loss: 0.0517
```

```
Mean surrogate loss: -0.0113
```

```
Mean entropy loss: 49.0651
```

```
Mean reward: -7.60
```

```
Mean episode length: 238.35
```

```
Episode_Reward/track_lin_vel_xy_exp: 0.0894
```

```
Episode_Reward/track_ang_vel_z_exp: 0.0610
```

```
Episode_Reward/lin_vel_z_l2: 0.0000
```

```
Episode_Reward/ang_vel_xy_l2: -0.0299
```

```
Episode_Reward/dof_torques_l2: -0.0004
```

```
Episode_Reward/dof_acc_l2: -0.0320
```

```
Episode_Reward/action_rate_l2: -0.0929
```

```
Episode_Reward/feet_air_time: 0.0011
```

```
Episode_Reward/flat_orientation_l2: -0.0098
```

```
Episode_Reward/dof_pos_limits: -0.0134
```

```
Episode_Reward/termination_penalty: -0.2000
```

```
Episode_Reward/feet_slide: -0.0161
```

```
Episode_Reward/joint_deviation_arms: -0.0259
```

```
Episode_Reward/joint_deviation_legs: -0.0163
```

```
Episode_Reward/base_height_l2: -0.0895
```

```
Curriculum/terrain_levels: 0.0364
```

```
Metrics/base_velocity/error_vel_xy: 0.2766
```

```
Metrics/base_velocity/error_vel_yaw: 0.9506
```

```
Episode_Termination/time_out: 0.0000
```

```
Episode_Termination/base_contact: 1.0000
```

```
-----  
Total timesteps: 11403264
```

```
Iteration time: 3.83s
```

```
Time elapsed: 00:14:59
```

```
ETA: 06:12:36
```

Second Attempt Logs

```
#####  
Learning iteration 282/6000
```

```
Computation: 11152 steps/s (collection: 4.158s, learning 0.250s)
```

```
Mean action noise std: 1.07
```

```
Mean value function loss: 0.0368
```

```
Mean surrogate loss: -0.0114
```

```
Mean entropy loss: 50.2552
```

```
Mean reward: -10.21
```

```
Mean episode length: 750.16
```

```
Episode_Reward/track_lin_vel_xy_exp: 0.3411
```

```
Episode_Reward/track_ang_vel_z_exp: 0.2089
```

```
Episode_Reward/lin_vel_z_l2: 0.0000
```

```
Episode_Reward/ang_vel_xy_l2: -0.0867
```

```
Episode_Reward/dof_torques_l2: -0.0013
```

```
Episode_Reward/dof_acc_l2: -0.1087
```

```
Episode_Reward/action_rate_l2: -0.3294
```

```
Episode_Reward/feet_air_time: 0.0029
```

```
Episode_Reward/flat_orientation_l2: -0.0153
```

```
Episode_Reward/dof_pos_limits: -0.0440
```

```
Episode_Reward/termination_penalty: -0.1010
```

```
Episode_Reward/feet_slide: -0.0585
```

```
Episode_Reward/joint_deviation_arms: -0.0867
```

```
Episode_Reward/joint_deviation_legs: -0.0571
```

```
Episode_Reward/base_height_l2: -0.1781
```

```
Curriculum/terrain_levels: 0.7601
```

```
Metrics/base_velocity/error_vel_xy: 0.7257
```

```
Metrics/base_velocity/error_vel_yaw: 2.9393
```

```
Episode_Termination/time_out: 0.3778
```

```
Episode_Termination/base_contact: 0.6222
```

```
-----  
Total timesteps: 13910016
```

```
Iteration time: 4.41s
```

```
Time elapsed: 00:18:39
```

```
ETA: 06:16:58
```

Second Attempt Logs

```
#####
Learning iteration 875/6000

      Computation: 11307 steps/s (collection: 4.100s, learning 0.246s)
      Mean action noise std: 1.31
      Mean value_function loss: 0.0259
      Mean surrogate loss: -0.0121
      Mean entropy loss: 53.5018
      Mean reward: -7.55
      Mean episode length: 924.87
      Episode_Reward/track_lin_vel_xy_exp: 0.6515
      Episode_Reward/track_ang_vel_z_exp: 0.4285
      Episode_Reward/lin_vel_z_l2: 0.0000
      Episode_Reward/ang_vel_xy_l2: -0.0607
      Episode_Reward/dof_torques_l2: -0.0017
      Episode_Reward/dof_acc_l2: -0.1426
      Episode_Reward/action_rate_l2: -0.6159
      Episode_Reward/feet_air_time: 0.0039
      Episode_Reward/flat_orientation_l2: -0.0103
      Episode_Reward/dof_pos_limits: -0.0396
      Episode_Reward/termination_penalty: -0.0465
      Episode_Reward/feet_slide: -0.0584
      Episode_Reward/joint_deviation_arms: -0.1292
      Episode_Reward/joint_deviation_legs: -0.0539
      Episode_Reward/base_height_l2: -0.3445
      Curriculum/terrain_levels: 5.7917
      Metrics/base_velocity/error_vel_xy: 0.5364
      Metrics/base_velocity/error_vel_yaw: 2.1876
      Episode_Termination/time_out: 0.8240
      Episode_Termination/base_contact: 0.1769
-----
      Total timesteps: 43057152
      Iteration time: 4.35s
      Time elapsed: 00:59:05
      ETA: 05:45:43
```

Second Attempt Logs

```
#####  
Learning iteration 1003/6000
```

```
Computation: 12163 steps/s (collection: 3.818s, learning 0.223s)
```

```
Mean action noise std: 1.35
```

```
Mean value_function loss: 0.0254
```

```
Mean surrogate loss: -0.0110
```

```
Mean entropy loss: 54.1137
```

```
Mean reward: -6.94
```

```
Mean episode length: 891.25
```

```
Episode_Reward/track_lin_vel_xy_exp: 0.6491
```

```
Episode_Reward/track_ang_vel_z_exp: 0.4385
```

```
Episode_Reward/lin_vel_z_l2: 0.0000
```

```
Episode_Reward/ang_vel_xy_l2: -0.0560
```

```
Episode_Reward/dof_torques_l2: -0.0016
```

```
Episode_Reward/dof_acc_l2: -0.1361
```

```
Episode_Reward/action_rate_l2: -0.6167
```

```
Episode_Reward/feet_air_time: 0.0031
```

```
Episode_Reward/flat_orientation_l2: -0.0104
```

```
Episode_Reward/dof_pos_limits: -0.0331
```

```
Episode_Reward/termination_penalty: -0.0458
```

```
Episode_Reward/feet_slide: -0.0572
```

```
Episode_Reward/joint_deviation_arms: -0.1302
```

```
Episode_Reward/joint_deviation_legs: -0.0533
```

```
Episode_Reward/base_height_l2: -0.3114
```

```
Curriculum/terrain_levels: 5.7107
```

```
Metrics/base_velocity/error_vel_xy: 0.4875
```

```
Metrics/base_velocity/error_vel_yaw: 1.9835
```

```
Episode_Termination/time_out: 0.8448
```

```
Episode_Termination/base_contact: 0.1552
```

```
-----  
Total timesteps: 49348608
```

```
Iteration time: 4.04s
```

```
Time elapsed: 01:07:54
```

```
ETA: 05:37:57
```

Second Attempt Logs

```
#####
Learning iteration 1527/6000

      Computation: 11830 steps/s (collection: 3.913s, learning 0.242s)
      Mean action noise std: 1.44
      Mean value function loss: 0.0244
      Mean surrogate loss: -0.0113
      Mean entropy loss: 55.3048
      Mean reward: -7.22
      Mean episode length: 901.66
Episode_Reward/track_lin_vel_xy_exp: 0.6361
Episode_Reward/track_ang_vel_z_exp: 0.4540
Episode_Reward/lin_vel_z_l2: 0.0000
Episode_Reward/ang_vel_xy_l2: -0.0521
Episode_Reward/dof_torques_l2: -0.0016
Episode_Reward/dof_acc_l2: -0.1330
Episode_Reward/action_rate_l2: -0.6571
Episode_Reward/feet_air_time: 0.0040
Episode_Reward/flat_orientation_l2: -0.0147
Episode_Reward/dof_pos_limits: -0.0298
Episode_Reward/termination_penalty: -0.0507
Episode_Reward/feet_slide: -0.0549
Episode_Reward/joint_deviation_arms: -0.1357
Episode_Reward/joint_deviation_legs: -0.0515
Episode_Reward/base_height_l2: -0.3064
Curriculum/terrain_levels: 5.8166
Metrics/base_velocity/error_vel_xy: 0.4626
Metrics/base_velocity/error_vel_yaw: 1.7943
Episode_Termination/time_out: 0.8300
Episode_Termination/base_contact: 0.1700
-----
Total timesteps: 75104256
Iteration time: 4.15s
Time elapsed: 01:43:07
ETA: 05:01:52
```

Seems like the training has plateaued. The mean reward stays around -7 to -8.



Second Attempt Logs

```
#####  
Learning iteration 2003/6000
```

```
Computation: 11941 steps/s (collection: 3.868s, learning 0.248s)
```

```
Mean action noise std: 1.47
```

```
Mean value function loss: 0.0240
```

```
Mean surrogate loss: -0.0099
```

```
Mean entropy loss: 55.5843
```

```
Mean reward: -6.26
```

```
Mean episode length: 890.80
```

```
Episode_Reward/track_lin_vel_xy_exp: 0.6854
```

```
Episode_Reward/track_ang_vel_z_exp: 0.5127
```

```
Episode_Reward/lin_vel_z_l2: 0.0000
```

```
Episode_Reward/ang_vel_xy_l2: -0.0491
```

```
Episode_Reward/dof_torques_l2: -0.0016
```

```
Episode_Reward/dof_acc_l2: -0.1343
```

```
Episode_Reward/action_rate_l2: -0.7160
```

```
Episode_Reward/feet_air_time: 0.0043
```

```
Episode_Reward/flat_orientation_l2: -0.0136
```

```
Episode_Reward/dof_pos_limits: -0.0258
```

```
Episode_Reward/termination_penalty: -0.0307
```

```
Episode_Reward/feet_slide: -0.0542
```

```
Episode_Reward/joint_deviation_arms: -0.1432
```

```
Episode_Reward/joint_deviation_legs: -0.0487
```

```
Episode_Reward/base_height_l2: -0.2877
```

```
Curriculum/terrain_levels: 5.7957
```

```
Metrics/base_velocity/error_vel_xy: 0.4557
```

```
Metrics/base_velocity/error_vel_yaw: 1.7361
```

```
Episode_Termination/time_out: 0.8217
```

```
Episode_Termination/base_contact: 0.1783
```

```
-----  
Total timesteps: 98500608
```

```
Iteration time: 4.12s
```

```
Time elapsed: 02:15:06
```

```
ETA: 04:29:28
```

Mean reward seems to decrease
more towards -6

Second Attempt Logs

```
#####  
Learning iteration 3007/6000
```

```
Computation: 11247 steps/s (collection: 4.137s, learning 0.233s)
```

```
Mean action noise std: 1.52
```

```
Mean value_function loss: 0.0253
```

```
Mean surrogate loss: -0.0118
```

```
Mean entropy loss: 55.5383
```

```
Mean reward: -4.87
```

```
Mean episode length: 937.17
```

```
Episode_Reward/track_lin_vel_xy_exp: 0.7432
```

```
Episode_Reward/track_ang_vel_z_exp: 0.6311
```

```
Episode_Reward/lin_vel_z_l2: 0.0000
```

```
Episode_Reward/ang_vel_xy_l2: -0.0497
```

```
Episode_Reward/dof_torques_l2: -0.0017
```

```
Episode_Reward/dof_acc_l2: -0.1455
```

```
Episode_Reward/action_rate_l2: -0.7966
```

```
Episode_Reward/feet_air_time: 0.0045
```

```
Episode_Reward/flat_orientation_l2: -0.0179
```

```
Episode_Reward/dof_pos_limits: -0.0248
```

```
Episode_Reward/termination_penalty: -0.0174
```

```
Episode_Reward/feet_slide: -0.0576
```

```
Episode_Reward/joint_deviation_arms: -0.1606
```

```
Episode_Reward/joint_deviation_legs: -0.0504
```

```
Episode_Reward/base_height_l2: -0.2638
```

```
Curriculum/terrain_levels: 5.8833
```

```
Metrics/base_velocity/error_vel_xy: 0.4750
```

```
Metrics/base_velocity/error_vel_yaw: 1.5673
```

```
Episode_Termination/time_out: 0.8443
```

```
Episode_Termination/base_contact: 0.1557
```

```
-----  
Total timesteps: 147849216
```

```
Iteration time: 4.37s
```

```
Time elapsed: 03:30:42
```

```
ETA: 03:29:39
```

Mean reward ranges from -4 to -6 now, leading closer to -4.

Second Attempt Logs

```
#####
Learning iteration 3509/6000

      Computation: 12436 steps/s (collection: 3.713s, learning 0.240s)
      Mean action noise std: 1.51
      Mean value_function loss: 0.0229
      Mean surrogate loss: -0.0102
      Mean entropy loss: 54.7581
      Mean reward: -3.31
      Mean episode length: 965.17
Episode_Reward/track_lin_vel_xy_exp: 0.7410
Episode_Reward/track_ang_vel_z_exp: 0.6811
Episode_Reward/lin_vel_z_l2: 0.0000
Episode_Reward/ang_vel_xy_l2: -0.0462
Episode_Reward/dof_torques_l2: -0.0017
Episode_Reward/dof_acc_l2: -0.1397
Episode_Reward/action_rate_l2: -0.7846
Episode_Reward/feet_air_time: 0.0052
Episode_Reward/flat_orientation_l2: -0.0263
Episode_Reward/dof_pos_limits: -0.0236
Episode_Reward/termination_penalty: -0.0229
Episode_Reward/feet_slide: -0.0532
Episode_Reward/joint_deviation_arms: -0.1579
Episode_Reward/joint_deviation_legs: -0.0493
Episode_Reward/base_height_l2: -0.3281
Curriculum/terrain_levels: 5.7384
Metrics/base_velocity/error_vel_xy: 0.4675
Metrics/base_velocity/error_vel_yaw: 1.4374
Episode_Termination/time_out: 0.9009
Episode_Termination/base_contact: 0.0991
-----
      Total timesteps: 172523520
      Iteration time: 3.95s
      Time elapsed: 04:03:35
      ETA: 02:52:52
```

Mean reward ranges from -2 to
-4 now.

Second Attempt Logs

```
#####  
Learning iteration 4507/6000
```

```
    Computation: 11793 steps/s (collection: 3.921s, learning 0.246s)
```

```
    Mean action noise std: 1.50
```

```
    Mean value_function loss: 0.0175
```

```
    Mean surrogate loss: -0.0119
```

```
    Mean entropy loss: 53.4491
```

```
    Mean reward: -1.06
```

```
    Mean episode length: 932.74
```

```
Episode_Reward/track_lin_vel_xy_exp: 0.7158
```

```
Episode_Reward/track_ang_vel_z_exp: 0.7133
```

```
    Episode_Reward/lin_vel_z_l2: 0.0000
```

```
    Episode_Reward/ang_vel_xy_l2: -0.0381
```

```
    Episode_Reward/dof_torques_l2: -0.0017
```

```
    Episode_Reward/dof_acc_l2: -0.1240
```

```
    Episode_Reward/action_rate_l2: -0.7117
```

```
    Episode_Reward/feet_air_time: 0.0050
```

```
Episode_Reward/flat_orientation_l2: -0.0242
```

```
    Episode_Reward/dof_pos_limits: -0.0191
```

```
Episode_Reward/termination_penalty: -0.0347
```

```
    Episode_Reward/feet_slide: -0.0456
```

```
Episode_Reward/joint_deviation_arms: -0.1447
```

```
Episode_Reward/joint_deviation_legs: -0.0388
```

```
    Episode_Reward/base_height_l2: -0.3220
```

```
    Curriculum/terrain_levels: 5.8682
```

```
Metrics/base_velocity/error_vel_xy: 0.3957
```

```
Metrics/base_velocity/error_vel_yaw: 1.1578
```

```
    Episode_Termination/time_out: 0.9080
```

```
    Episode_Termination/base_contact: 0.0920
```

```
-----  
Total timesteps: 221577216
```

```
Iteration time: 4.17s
```

```
Time elapsed: 05:16:42
```

```
ETA: 01:44:53
```

Second Attempt Logs

```
#####  
Learning iteration 5999/6000
```

```
Computation: 7690 steps/s (collection: 6.005s, learning 0.386s)
```

```
Mean action noise std: 1.47
```

```
Mean value_function loss: 0.0205
```

```
Mean surrogate loss: -0.0098
```

```
Mean entropy loss: 51.8108
```

```
Mean reward: 2.84
```

```
Mean episode length: 950.56
```

```
Episode_Reward/track_lin_vel_xy_exp: 0.7773
```

```
Episode_Reward/track_ang_vel_z_exp: 0.8345
```

```
Episode_Reward/lin_vel_z_l2: 0.0000
```

```
Episode_Reward/ang_vel_xy_l2: -0.0350
```

```
Episode_Reward/dof_torques_l2: -0.0030
```

```
Episode_Reward/dof_acc_l2: -0.1134
```

```
Episode_Reward/action_rate_l2: -0.7109
```

```
Episode_Reward/feet_air_time: 0.0051
```

```
Episode_Reward/flat_orientation_l2: -0.0285
```

```
Episode_Reward/dof_pos_limits: -0.0154
```

```
Episode_Reward/termination_penalty: -0.0208
```

```
Episode_Reward/feet_slide: -0.0433
```

```
Episode_Reward/joint_deviation_arms: -0.1457
```

```
Episode_Reward/joint_deviation_legs: -0.0391
```

```
Episode_Reward/base_height_l2: -0.3121
```

```
Curriculum/terrain_levels: 5.7834
```

```
Metrics/base_velocity/error_vel_xy: 0.3711
```

```
Metrics/base_velocity/error_vel_yaw: 1.0501
```

```
Episode_Termination/time_out: 0.9115
```

```
Episode_Termination/base_contact: 0.0885
```

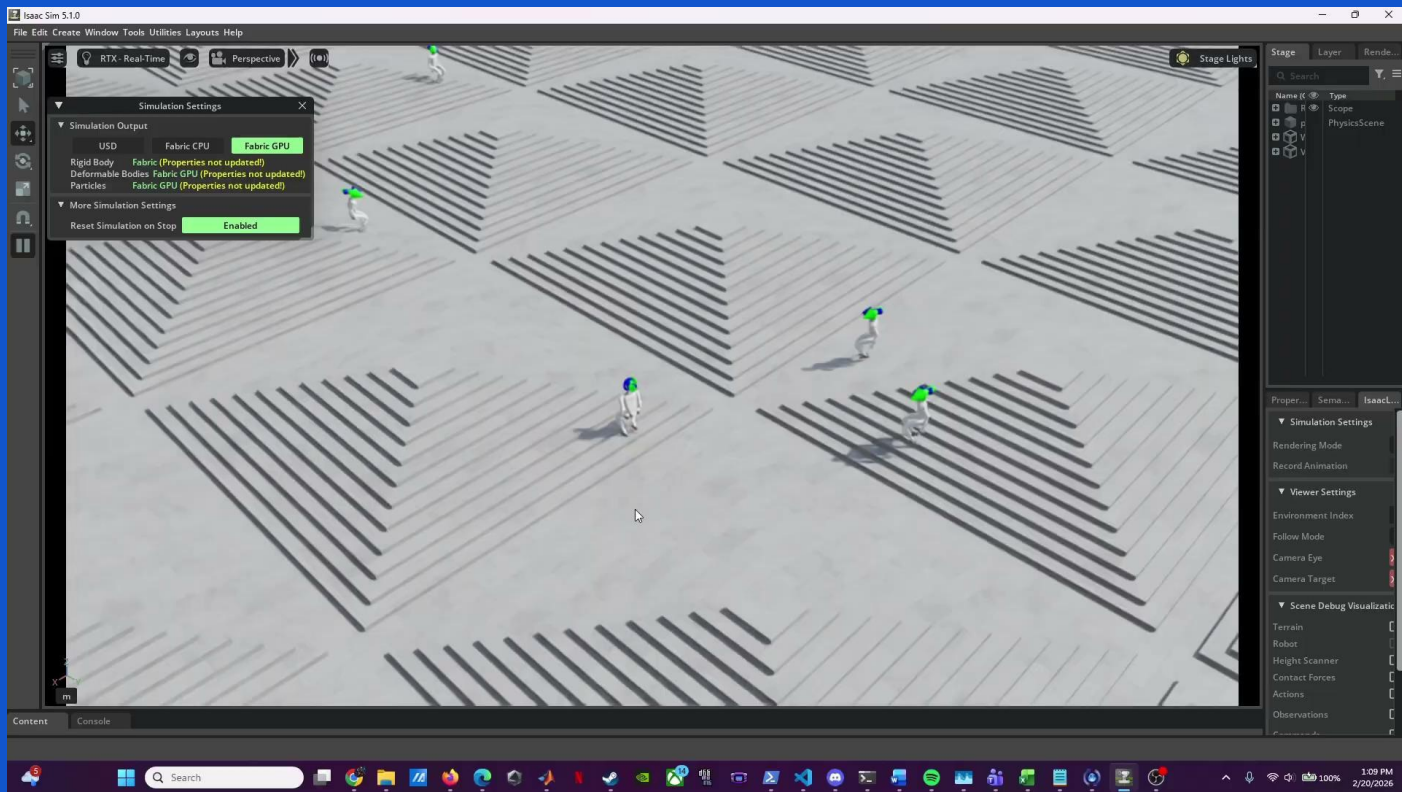
```
-----  
Total timesteps: 294912000
```

```
Iteration time: 6.39s
```

```
Time elapsed: 07:11:16
```

```
ETA: 00:00:04
```

```
.\isaacsim.bat -p scripts/reinforcement_learning/rsl_rl/play.py --task  
Isaac-Velocity-Stairs-G1-Play-v0 --checkpoint  
"C:\Users\Mario\IsaacLab\logs\rsl_rl\g1_stairs\2026-02-19_13-19-06\model_5999.pt"
```



Analysis on Second Attempt

- The g1 robot now walks in a squatting position with its legs inwards.
 - There may be an issue with height constraints (rewards based on how high the robot can stand). This might cause the robot to remain in that squatting position to prevent penalties.

Third Attempt

- Added a `base_height` reward into the code (`stair_env_cfg.py`) so that the robot is forced to maintain a specific height.
 - After fine tuning this attempt, it caused the robot to not be able to walk (assuming it's because it didn't want to get penalized for going under this base height).

```
45     base_height_l2 = RewTerm(  
46         func=mdp.base_height_l2,  
47         weight=-1.5,  
48         params={"target_height": 0.80, "asset_cfg": SceneEntityCfg("robot")},  
49     )  
50
```

Next Steps

- Need to perfect the g1 robot's walking stance.
- Need to learn more about environment/terrain customization.
 - For example, it would be even easier if I made a simple terrain in which its one straight, forcing the robot to walk up a normal flight of stairs instead of pyramids.
- Find way to train G1 to open doors
 - Main things we want g1 to do: open doors, walk up stairs, enter rooms, pick up items.

Sources

- <https://www.youtube.com/watch?v=S-MXmMoV3zQ>
- https://github.com/isaac-sim/IsaacLab/blob/main/source/isaacsim_tasks/isaacsim_tasks/manager_based/locomotion/velocity_controller/g1/agents/rsl_rl_ppo_cfg.py
- <https://isaac-sim.github.io/IsaacLab/main/index.html>

Note Log Names

- Attempt 1: 2026-02-17_13-20-13
- Attempt 2: 2026-02-19_13-19-06